

Simon Chen

347-322-4133 | shangminch@gmail.com | simon-chen.com | linkedin.com/in/shangmin-chen | github.com/shangmin-chen

EDUCATION

Boston University

Boston, MA

Bachelor of Arts in Computer Science

Graduated: Spring 2026

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Distributed Systems, Databases, Machine Learning, Data Science.

EXPERIENCE

Software Engineering Intern

February 2026 – June 2026

RESET Standard

Shanghai, China

- Engineered asynchronous data pipelines with RabbitMQ, Redis, and Sidekiq to process long-running ingestion jobs in the background, decoupling external API calls from user-facing requests.
- Doubled the platform's supported environmental data categories (from 2 to 4) by independently implementing end-to-end water and waste data integration, developing Rails backend ingestion pipelines, PostgreSQL schemas, and client-facing React/Vite/Tailwind dashboards.
- Redesigned project-device mapping workflows in response to client requests, refactoring legacy forms to streamline onboarding of environmental monitoring devices.

Lead Product Engineer

December 2025 – Present

EZ Esports

New York, NY

- Designed a PostgreSQL schema on Supabase using Drizzle ORM, supporting 500+ matches across 20+ teams, and built RESTful endpoints for player statistics, standings, and historical data.
- Migrated an esports league platform from Google Sites to Next.js with a Supabase backend, replacing manual content updates with live scoring and automated season management.
- Cut page load time by 75% (from 3.2s to 0.8s) by implementing edge caching on Cloudflare, route-based code splitting, and asset optimization for the Next.js frontend.

PROJECTS

Persephone | *Python, Cython/C++17, Rust, Modal, FAISS*

May 2026

- Cut backtest and parameter-sweep iteration from ~14 hours on laptop CPUs to ~15 minutes by moving tick backtests and sweeps to Modal NVIDIA GPU workers with FAISS GPU search and CuPy-vectorized math.
- Replaced KDE inference with a GBM theo layer, cutting model query latency from ~1.5ms to ~20μs per query (~75x) behind an unchanged live serving interface.
- Sped up 155K-row neighbor search from 758μs to 154μs p50 by caching FAISS IVF indexes and pinning FAISS threads to avoid OpenMP oversubscription.

Whisperrr | *Java Spring Boot, Python FastAPI, React, PostgreSQL / Tailwind CSS*

September 2025

- Built full-stack transcription platform processing audio at 32x realtime speed, supporting 99+ languages across multiple audio formats with real-time results
- Designed three-tier stateless architecture with HTTP connection pooling to reduce request latency by 70%, improving reliability for files over 25MB from 60% to 100% success rate

Hermes Letters | *Next.js, TypeScript, Supabase, Drizzle ORM, PostgreSQL*

June 2026

- Ensured consistent letter claims via atomic SQL updates and secured sessions in XSS-resistant httpOnly cookies.
- Derived user connections at query time from kept-letter records using indexed lookups, avoiding a separate relationship table and associated database sync issues.

TECHNICAL SKILLS

Languages: Python, SQL, C++17, Cython, Rust, TypeScript, JavaScript, Java, Ruby

Frameworks & Libraries: FAISS, CuPy, FastAPI, Django, React, Next.js, Node.js, Vite, Spring Boot, Ruby on Rails, React Native, Tailwind CSS

Databases & Infrastructure: Modal, PostgreSQL, SQLite, Docker, Redis, RabbitMQ, Sidekiq, Supabase, Cloudflare, Nginx, OpenMP